

PREPARE-THRIVE-SUCCEED

IPU NOTES

B.TECH CSE

3RD SEM



SNAPED
COMMUNITY

COMPUTATIONAL METHODS
UNIT-1

to read today

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Unit – I

Syllabus

Review of Taylor Series, Rolle 's Theorem and Mean Value Theorem, Approximations and Errors in numerical computations, Data representation and computer arithmetic, Loss of significance in computation.

Location of roots of equation: Bisection method (convergence analysis and implementation), Newton Method (convergence analysis and implementation), Secant Method (convergence analysis and implementation).

Unconstrained one variable function minimization by Fibonacci search, Golden Section Search and Newton's method. Multivariate function minimization by the method of steepest descent, Nelder-Mead Algorithm.

- **Taylor's Theorem**

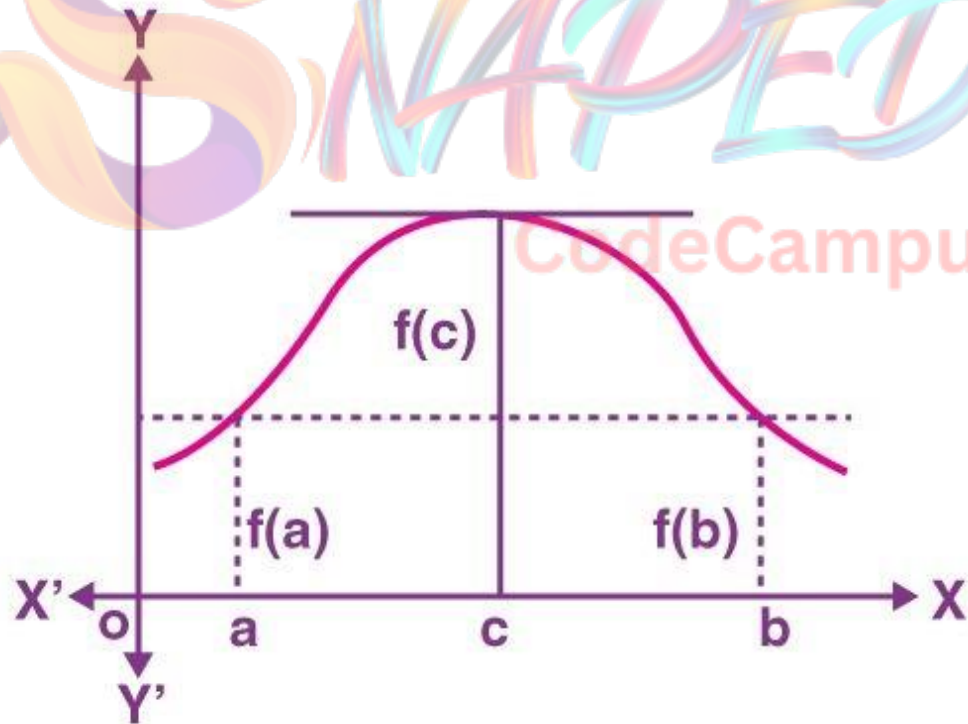
If $f(x)$ is continuous and differential, the $f(x)$ can be expanded in a series as

$$f(x + h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \dots$$

CodeCampus

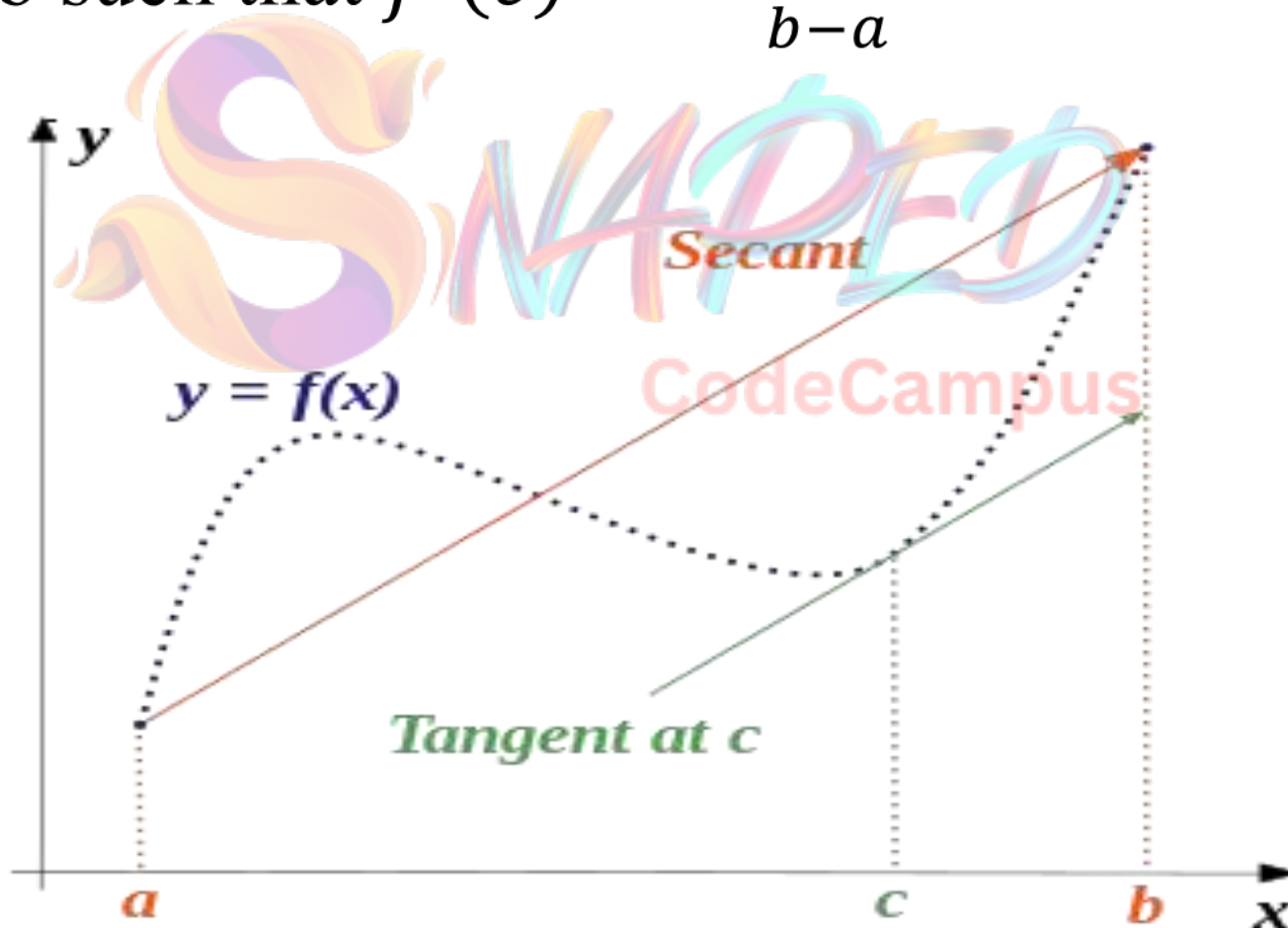
Rolle's Theorem

If $f(x)$ is continuous on $[a, b]$, differentiable in (a, b) and $f(a) = f(b)$, then there exist a point c between a and b such that $f'(c) = 0$



- # Mean Value Theorem

If $f(x)$ is continuous on $[a, b]$ and differentiable in (a, b) then there exist a point c between a and b such that $f'(c) = \frac{f(b) - f(a)}{b - a}$



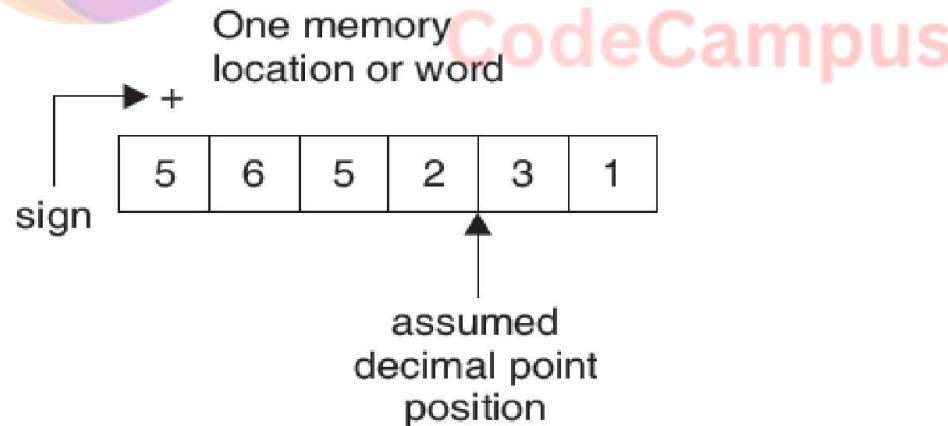
Floating Point Arithmetic

There are two types of arithmetic operations available in a computer:

- (i) **Integer arithmetic** -- used mainly in counting and as subscripts.
- (ii) **Real or floating point arithmetic**: - uses numbers with fractional parts as operands.

Computers are usually designed such that each location, called **word**, in memory stores only a finite number of digits. Consequently, all operands in arithmetic operations have only a finite number of digits.

Assume: computer having memory in which each location can store 6 digits.



A memory location storing number 5652.31

Shifting of the mantissa to the left until its most significant digit is non-zero is called **normalization**.

Arithmetic operations with Floating Point numbers:

Addition and subtraction:

The exponents of the two numbers must be made equal (Larger one) and the Mantissa shifted appropriately.

Ex1. Add the following

(i).4146 E 7 & .3433 E 7 (ii) .2546 E 5 & .5434 E 7 (iii) .9432 E - 4 & .5452 E - 3

Solution: (i) Since exponent of both numbers is same, we can add simply

```
.4146 E 7
.3433 E 7
-----
.7579 E 7
```

(ii) Since exponents of both the numbers are different, first we make them equal (7) and write mantissa of second accordingly.

```
.0025 E 7
.5434 E 7
-----
.5459 E 7
```

(iii) Since exponent is not equal, we make it equal (-3 larger than -4)

```
.0943 E -3
.5452 E -3
-----
.6395 E -3
```

Note: If the result of an arithmetic operation gives a number smaller than $.1000 \text{ E } -99$ then it is called an **underflow** condition. Similarly, any result greater than $.9999 \text{ E } 99$ leads to an **overflow** condition.

Multiplication and division

Multiplication:

Multiply the mantissas and add the exponents.

Normalize the resulting mantissa

Example: (i) $.5543 \text{ E } 12 \times .4111 \text{ E } -15$

Ans $.2278 \text{ E } -3$

(ii) $.1111 \text{ E } 10 \times .1234 \text{ E } 15$

Ans $.1370 \text{ E } 24$

Division:

The mantissa of the numerator is divided by that of the denominator. The denominator exponent is subtracted from the numerator exponent.

Example: (i) $.9998 \text{ E } 1 \div .1000 \text{ E } -99$

Ans $.9998 \text{ E } 101$, Hence the result **overflows**.

(ii) $.9998 \text{ E } -5 \div .1000 \text{ E } 98$

Ans $.9998 \text{ E } -104$, Hence the result **underflows**.

(iii) $.1000 \text{ E } 5 \div .9999 \text{ E } 3$

Ans. $.1000 \text{ E } 2$.

Associative and Distributive laws in FP arithmetic:

The associative and distributive laws are, in general, not followed by floating point arithmetic. **Examples:**

$$a = 6.31 \times 10^1, \quad b = 4.24 \times 10^0, \quad c = 2.47 \times 10^{-1},$$

$$\begin{aligned}(a + b) + c &= (6.31 \times 10^1 + 0.424 \times 10^1) + 2.47 \times 10^{-1} \\ &= 6.73 \times 10^1 + 0.0247 \times 10^1 \\ &= 6.75 \times 10^1\end{aligned}$$

$$\begin{aligned}a + (b + c) &= 6.31 \times 10^1 + (4.24 \times 10^0 + 0.247 \times 10^0) \\ &= 6.31 \times 10^1 + 4.49 \times 10^0 \\ &= 6.31 \times 10^1 + 4.49 \times 10^0 \\ &= 6.31 \times 10^1 + 0.449 \times 10^1 \\ &= 6.76 \times 10^1\end{aligned}$$

Similarly, we can show that distributive laws are not followed in floating point arithmetic.

Errors

Machine epsilon

A computer has a finite word length, so only a fixed number of digits is stored and used during computation. Hence, even in storing an exact decimal number in its converted form in the computer memory, an error is introduced. This error is machine dependant and is called **machine epsilon**.

$$\text{Error} = \text{True value} - \text{Approximate value}$$

In order to determine the accuracy in an approximate solution to a problem, either we find the bound of the

$$\text{Relative error} = \frac{|Error|}{|Truevalue|}$$

or the **Absolute error** = $|Error|$

Inherent Error: A quantity which is already present in the statement of the problem before its solution.

Two reasons: (i) The simplified assumptions in the mathematical formulation of the problem (ii) error in the physical measurements of the parameters of the problem.

For example: Area of a rectangle. Scale has error

Errors

Rules of rounding off:

Round - off error: The round off error is the quantity which must be added to the finite representation of a computed number in order to make it the true representation of that number.

Chopping error: All the digits after required number of digits discarded.

Thus, if x is the computed number given by

$$x = .d_1 d_2 \dots d_t d_{t+1} \dots \times 10^r$$

then the relative error for t -digit mantissa standard form representation of x becomes

$$\frac{|x - fl(x)|}{|x|} \leq \begin{cases} 10^{1-t}, & \text{for chopping} \\ \frac{1}{2} 10^{1-t}, & \text{for rounding} \end{cases}$$

Thus, the bound on the relative error of a floating – point number is reduced by half when rounding is used than chopping. We write

$fl(x) = x(1 + \delta)$, where δ is some number depending on x called the relative round –off error in $fl(x)$. ($fl(x)$ is floating point representation of x).

The number δ is called the **machine epsilon** and is denoted by EPS . Thus,

$$|\delta| = EPS = \begin{cases} 10^{1-t}, & \text{for chopping} \\ \frac{1}{2} 10^{1-t}, & \text{for rounding} \end{cases}$$

Errors

Truncation error: The truncation error is the quantity which must be added to the true representation of the quantity in order the result be exactly equal to the quantity we are seeking to generate.

This error is a result of the appropriate formulae used which are generally based on truncated series. Taylor's series with a remainder is an invaluable tool in study of the truncation error. For example

Numerical Instability: Every arithmetic operation performed during computation, gives rise to some error, which once generated may decay or grow in subsequent calculations. In some cases error may grow so large as to make the computed result totally redundant and we shall call such a procedure, **numerically unstable**.

For example: Let $f(x) = 1/(1-x^2)$ and we want to calculate $f(x)$ for $x = 0.9$. The exact value of $f(x)$ for $x = 0.9$ correct to six digits is $f(x) = 0.526316 \times 10^1$. If x is approximated to $x^* = 0.900005$, i.e. an error is introduced in the sixth decimal place, we obtain the value $f(x^*) = 0.526341 \times 10^1$.

Thus, an error in the sixth place has caused an error in the fifth place in $f(x)$. Thus, when a number is divided by a very small number, there is loss of significant digits and magnification of errors in the result.

Assignment 1

1.
 - (i) Add $.31251 \times 10^5$ and $.69013 \times 10^{-2}$
 - (ii) Subtract $.1259 \times 10^3$ from $.2143 \times 10^2$
 - (iii) Multiply $.317 \times 10^6$ and $.693 \times 10^{-4}$
 - (iv) Divide $.7251 \times 10^7$ and $.5997 \times 10^{-2}$
2. Define Absolute error and relative error. An approximate value of π is given by 3.1428571 and its true value is 3.1415926. Find absolute and relative errors.
3. What is numerical instability? Explain with an example.
4. Find machine epsilon for chopping and rounding off for a floating point of 7 point mantissa.

Roots of equations

Algebraic (polynomial) and Transcendental equations:

$2x^3 + 3x^2 - 2x + 1 = 0$ is an algebraic equation.

Equations other than algebraic, are called transcendental and they involve trigonometric, logarithmic or exponential functions. E.g., $x - \cos x = 0$, $x \log x - 3 = 0$, $e^x - \cos x = 0$ etc.

Solution of an equation: A number α is called a soln of equation $f(x) = 0$ if $f(\alpha) = 0$. Such a soln α is a root or a zero of $f(x) = 0$. Geometrically, α is a value of x where $f(x)$ intersects x -axis.

Two types of solution methods:

- 1. Direct methods-** exact values of roots, all the roots at the same time
- 2. Iterative methods (Indirect methods)-** Based on the idea of successive approximations, only one root at a time, rarely exact soln.

Initial Approximations

Intermediate value theorem/property:

$f(x)$ – continuous on $[a,b]$,

$f(a)$ & $f(b)$ have opposite signs (i.e. $f(a).f(b)<0$),

$f(x) = 0$ has at least one real root in the interval (a, b)

We choose any value in the interval (a, b) . This value is called **initial approximation** of x and is denoted by x_0 .

When to stop the iterations

In numerical methods, most of times we find approximate values. In this case, it is important to see the level of accuracy. There are three types of problems:

- (1) Accuracy upto some specific place of decimal
- (2) Accuracy upto some specific significant digits
- (3) Nothing is specified.
- (4) Number of iterations is specified

In **case (1)**, we stop the iteration when the digits after decimal to required accuracy are repeated in two consecutive iterations. For example if we have to find a root correct to four decimal places, then we stop when two consecutive values after four places of decimal are repeated.

In **case (2)**, same as case (1), in place of decimal place, we consider significant digit.

In **case (3)**, We follow case (1) for at least 3 or 4 digits accuracy after decimal.

In **case (4)**, perform the number of iterations specified.

Bisection Method

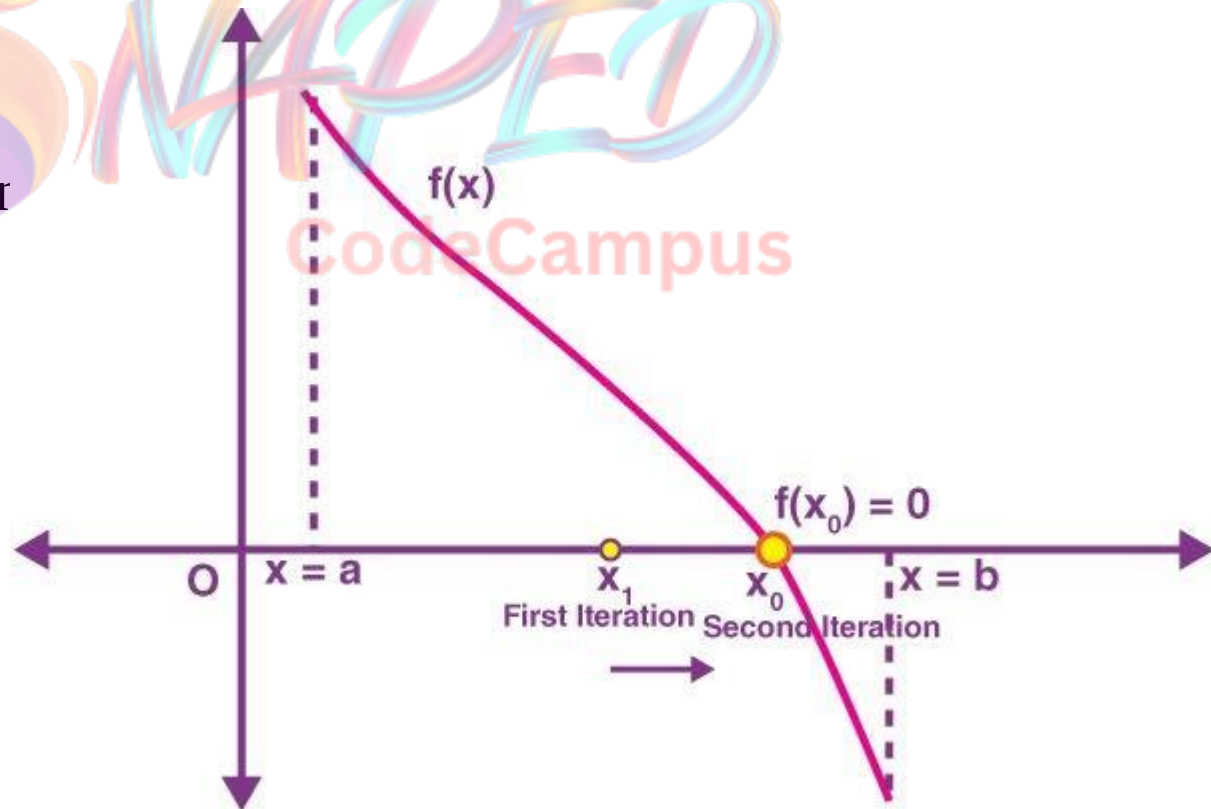
Solution of $f(x) = 0$: Let a, b are chosen from IVP. Then first approximation is given by

$$x_1 = (a + b)/2$$

If $f(x_1) = 0$, then x_1 is a root of $f(x) = 0$. Otherwise, the root lies between a and x_1 or x_1 and b according as $f(x_1)$ is positive or negative. Then we bisect the interval as before and continue the process until the root is found to desired accuracy.

Ex Find a root of the equation $x^3 - 4x - 9 = 0$, using the bisection method correct to three decimal places.

Ans. 2.706



Newton-Raphson's Method/Newton Method

This method is generally used to improve the result obtained by one of previous methods. Let x_0 be an approximate root of $f(x) = 0$ and let $x_1 = x_0 + h$ the correct root so that $f(x_1) = 0$.

Expanding $f(x_0 + h)$ by Taylor's series, we get

$$f(x_0) + hf'(x_0) + \frac{h^2}{2!} f''(x_0) + \dots = 0$$

Since h is small, neglecting h^2 and higher powers of h , we get

$$f(x_0) + hf'(x_0) = 0 \quad \text{or} \quad h = - \frac{f(x_0)}{f'(x_0)} \quad ($$

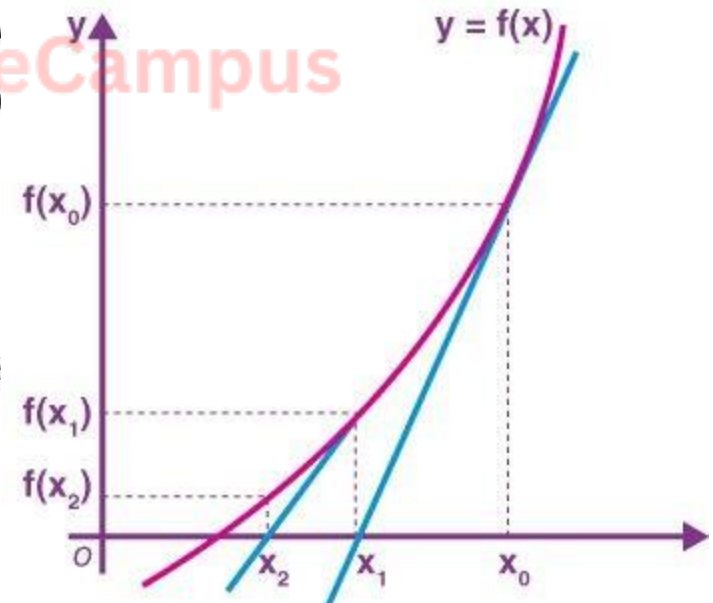
A better approximation than x_0 is therefore

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Successive approximations are given by x_2

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

which is the Newton-Raphson formula.



Example 1. Using Newton-Raphson method, find the real root of the equation $3x = \cos x + 1$ correct to four decimal places. Give computer program using 'C'.

Sol. Let

$$f(x) = 3x - \cos x - 1$$

Since

$$f(0) = -2 = (-)\text{ve};$$

$$f(1) = 1.4597 = (+)\text{ve}$$

$\therefore$ A root of $f(x) = 0$ lies between 0 and 1. It is nearer to 1. Let us take $x_0 = 0.6$.

Also,

$$f'(x) = 3 + \sin x$$

Newton's iteration formula gives,

$$\begin{aligned} x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ &= x_n - \frac{3x_n - \cos x_n - 1}{3 + \sin x_n} = \frac{x_n \sin x_n + \cos x_n + 1}{3 + \sin x_n} \end{aligned}$$

If $n = 0$, the first approximation x_1 is given by,

$$\begin{aligned} x_1 &= \frac{x_0 \sin x_0 + \cos x_0 + 1}{3 + \sin x_0} \\ &= \frac{0.6 \sin 0.6 + \cos 0.6 + 1}{3 + \sin 0.6} = .6071 \end{aligned}$$

If $n = 1$,

the second approximation is

$$\begin{aligned} x_2 &= \frac{x_1 \sin x_1 + \cos x_1 + 1}{3 + \sin x_1} \\ &= \frac{.6071 \sin (.6071) + \cos (.6071) + 1}{3 + \sin (.6071)} = 0.6071 \end{aligned}$$

Clearly $x_1 = x_2$. Hence the desired root is **0.6071**, correct to 4 decimal places.

Example2. Using Newton-Raphson's method, find a real root of the equation $x \log_{10} x - 1.2 = 0$ correct to five decimal places.

Solution: $f(1) = -1.2$, -ve and $f(3) = 0.2313$, +ve.

So a root of $f(x) = 0$ lies between 1 and 3.

Let us take $x_0 = 2$.

and $f'(x) = \log_{10} x + \log_{10} e = \log_{10} x + 0.43429$

Newton's iteration formula gives,

$$\begin{aligned} x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ &= x_n - \frac{x_n \log_{10} x_n - 1.2}{\log_{10} x_n + .43429} = \frac{.43429 x_n + 1.2}{\log_{10} x_n + .43429} \end{aligned}$$

Given $n = 0$, the first approximation is

$$x_1 = \frac{.43429 x_0 + 1.2}{\log_{10} 2 + .43429} = 2.81 \quad (\because)$$

Similarly, given $n = 1, 2, 3, 4$ in (33), we get

$$x_2 = 2.741, x_3 = 2.74064, x_4 = 2.74065, x_5 = 2.74065$$

Clearly, $x_4 = x_5$

Hence the required root is **2.74065**, correct to five decimal places.

Example3. Find cube root of 17 using Newton's iterative method correct to five decimal places. Or compute $(17)^{1/3}$ using Newton's method correct to four decimal places.



Secant Method

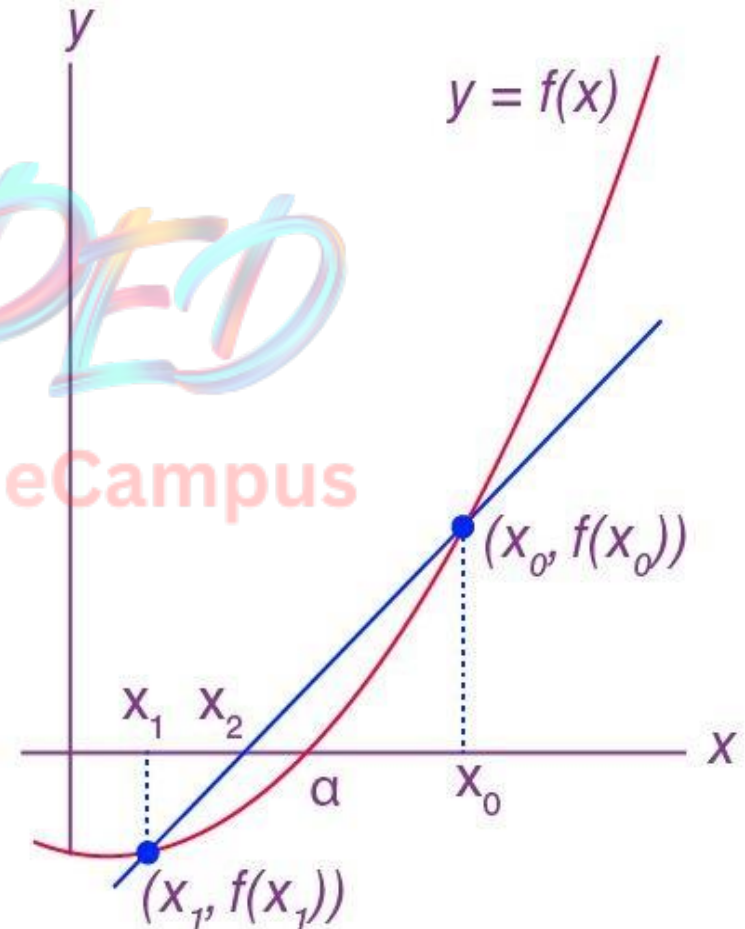
- **Solution of $f(x) = 0$:** Let a, b are chosen from IVP.

Writing x_0 and x_1 respectively for a & b and generalizing, we get

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

Or
$$x_{n+1} = \frac{x_{n-1}f(x_n) - x_nf(x_{n-1})}{f(x_n) - f(x_{n-1})}$$

Note that secant method uses two initial approximations.



Example1. A root of the equation $x^3 - 5x + 1 = 0$ lies in the interval $(0, 1)$. Perform four iterations of secant method. **Ans.** 0.201640

• **or**

Find a real root of the equation $x^3 - 5x + 1 = 0$ using secant method perform four iterations. **Ans.** 0.201640

• **or**

Find a real root of the equation $x^3 - 5x + 1 = 0$ using secant method correct to three decimal places. **Ans.** 0.201

• **or**

Find a real root of the equation $x^3 - 5x + 1 = 0$ using secant method with three significant digits. **Ans.** 0.201

Example2. Find the root of the equation $xe^x = \cos x$ using the secant method correct to four decimal places. **Ans.** 0.5177

Soln: $f(x) = \cos x - xe^x$. $f(0) = 1$, $f(1) = \cos 1 - e = -2.17798$
0.31467, 0.44673, 0.53171, 0.51690, 0.51775, 0.51776, etc.

Rate of convergence

The fastness of convergence in any method is represented by its rate of convergence.

An iterative method is said to be k th order convergent if k is the largest positive real number such that

$$\lim_{i \rightarrow \infty} \left| \frac{e_{i+1}}{e_i^k} \right| \leq A$$

where A is a non-zero finite number called asymptotic error constant .

Note: If e_{i+1}/e_i is almost constant, convergence is said to be **linear** i.e. slow. If e_{i+1}/e_i^p is nearly constant, convergence is said to be of **order p** i.e. faster

Rate of convergence

Bisection Method:



Rate of Convergence

Secant method: Let α be a root of $f(x) = 0$. Substitute $x_k = \alpha + \varepsilon_k$ in the formula

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$$

we obtain,

$$\varepsilon_{n+1} = \varepsilon_n - \frac{(\varepsilon_n - \varepsilon_{n-1})f(\alpha + \varepsilon_n)}{f(\alpha + \varepsilon_n) - f(\alpha + \varepsilon_{n-1})}$$

Expanding, in Taylor's series

$$\begin{aligned} \varepsilon_{n+1} &= \varepsilon_n - \frac{(\varepsilon_n - \varepsilon_{n-1})[f(\alpha) + \varepsilon_n f'(\alpha) + \frac{1}{2}\varepsilon_n^2 f''(\alpha) + \dots]}{(\varepsilon_n - \varepsilon_{n-1})f'(\alpha) + \frac{1}{2}(\varepsilon_n^2 - \varepsilon_{n-1}^2)f''(\alpha) + \dots} \\ &= \varepsilon_n - \frac{(\varepsilon_n - \varepsilon_{n-1})[f(\alpha) + \varepsilon_n f'(\alpha) + \frac{1}{2}\varepsilon_n^2 f''(\alpha) + \dots]}{(\varepsilon_n - \varepsilon_{n-1})f'(\alpha) + \frac{1}{2}(\varepsilon_n^2 - \varepsilon_{n-1}^2)f''(\alpha) + \dots} \end{aligned}$$

$$\begin{aligned}\varepsilon_{k+1} &= \varepsilon_k - \left[\varepsilon_k + \frac{1}{2} \varepsilon_k^2 \frac{f''(\alpha)}{f'(\alpha)} + \dots \right] \left[1 + \frac{1}{2} (\varepsilon_{k-1} + \varepsilon_k) \frac{f''(\alpha)}{f'(\alpha)} + \dots \right]^{-1} \\ &= \frac{1}{2} \frac{f''(\alpha)}{f'(\alpha)} \varepsilon_k \varepsilon_{k-1} + \text{higher powers of } \varepsilon_k\end{aligned}$$

$$\text{or } \varepsilon_{k+1} = C \varepsilon_k \varepsilon_{k-1} \quad (1)$$

where $C = \frac{1}{2} \frac{f''(\alpha)}{f'(\alpha)}$, and neglect higher powers of ε_n .

Now, We need a relation of the form $\varepsilon_{k+1} = A \varepsilon_k^p$ (2)

From (2), we have $\varepsilon_k = A \varepsilon_{k-1}^p$ or $\varepsilon_{k-1} = A^{-1/p} \varepsilon_k^{1/p}$

Substituting values in (1), we get $\varepsilon_k^p = C A^{-(1+1/p)} \varepsilon_k^{1+1/p}$ (3)

Comparing the powers on both sides, we get $p = 1 + 1/p$

which gives $p = \frac{1}{2}(1 \pm \sqrt{5})$. Neglecting minus sign, $p = 1.618$

Thus, convergence rate of secant method is 1.618.

Rate of Convergence

Newton-Raphson Method:

Let

$$x_n = \alpha + \varepsilon_n,$$

$$x_{n+1} = \alpha + \varepsilon_{n+1}$$

where α is a root of $f(x) = 0$. If we can prove that $\varepsilon_{n+1} = K \varepsilon_n^p$, where K is a constant and ε_n is the error involved at the n^{th} step, while finding the root by an iterative method, then the rate of convergence of the method is p .

We can now establish that Newton-Raphson method converges quadratically.

Let

$$x_n = \alpha + \varepsilon_n,$$

$$x_{n+1} = \alpha + \varepsilon_{n+1},$$

where α is a root of $f(x) = 0$ and ε_n is the error involved at the n^{th} step, while finding the root by Newton-Raphson formula (1). Then, Eq. (1) gives

$$\alpha + \varepsilon_{n+1} = \alpha + \varepsilon_n - \frac{f(\alpha + \varepsilon_n)}{f'(\alpha + \varepsilon_n)},$$

i.e.,

$$\varepsilon_{n+1} = \varepsilon_n - \frac{f(\alpha + \varepsilon_n)}{f'(\alpha + \varepsilon_n)} = \frac{\varepsilon_n f'(\alpha + \varepsilon_n) - f(\alpha + \varepsilon_n)}{f'(\alpha + \varepsilon_n)}$$

Using Taylor's expansion, we get

$$\varepsilon_{n+1} = \frac{1}{f'(\alpha) + \varepsilon_n f''(\alpha) + \dots} \{ \varepsilon_n [f'(\alpha) + \varepsilon_n f''(\alpha) + \dots] \}$$

$$- \left[f(\alpha) + \varepsilon_n f'(\alpha) + \frac{\varepsilon_n^2}{2} f''(\alpha) + \dots \right]$$

Since α is a root, $f(\alpha) = 0$. Therefore, the above expression simplifies to

$$\begin{aligned} \varepsilon_{n+1} &= \frac{\varepsilon_n^2}{2} f''(\alpha) \frac{1}{f'(\alpha) + \varepsilon_n f''(\alpha)} \\ &= \frac{\varepsilon_n^2}{2} \frac{f''(\alpha)}{f'(\alpha)} \left[1 + \varepsilon_n \frac{f''(\alpha)}{f'(\alpha)} \right]^{-1} \\ &= \frac{\varepsilon_n^2}{2} \frac{f''(\alpha)}{f'(\alpha)} \left[1 - \varepsilon_n \frac{f''(\alpha)}{f'(\alpha)} \right] \end{aligned}$$

or

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2} \frac{f''(\alpha)}{f'(\alpha)} + o(\varepsilon_n^3)$$

On neglecting terms of order (ε_n^3) and higher power, we obtain

$$\varepsilon_{n+1} = K(\varepsilon_n^2), \text{ where} \quad (3)$$

$$K = \frac{f''(\alpha)}{2f'(\alpha)} \quad (4)$$

It shows that Newton-Raphson method has second order convergence or converges quadratically.

Modified Newton - Raphson Method

Method for multiple roots.

If $f(x) = 0$ has a root α which is repeated m times, then $f(x)$ can be expressed as $f(x) = (x - \alpha)^m g(x)$ where $g(x)$ is a polynomial of degree $n-m$ (degree of $f(x)$ is n), then α is called a root with multiplicity m .

Modified Newton-Raphson method: To find a root of $f(x) = 0$ of multiplicity m , modified Newton-Raphson formula is:

$$x_{n+1} = x_n - m \frac{f(x_n)}{f'(x_n)}$$

Note: If α is a root of $f(x) = 0$ with multiplicity m , then it is also a root of $f'(x) = 0$ with multiplicity $m-1$, of $f''(x) = 0$ with multiplicity $m-2$ and so on. Hence if initial approximation x_0 is sufficiently close to the root, then the expressions

$$x_0 - m \frac{f(x_0)}{f'(x_0)}, x_0 - (m-1) \frac{f'(x_0)}{f''(x_0)}, x_0 - (m-2) \frac{f''(x_0)}{f'''(x_0)}, \dots$$

will have the same value

Example1: Find the double root of the equation

$$x^3 - x^2 - x + 1 = 0.$$

Sol. Let

$$f(x) = x^3 - x^2 - x + 1$$

so that

$$f'(x) = 3x^2 - 2x - 1$$

$$f''(x) = 6x - 2$$

Starting with $x_0 = 0.9$, we have

$$x_0 - 2 \frac{f(x_0)}{f'(x_0)} = .9 - \frac{2 \times .019}{(-.37)} = 1.003$$

and

$$x_0 - (2 - 1) \frac{f'(x_0)}{f''(x_0)} = .9 - \frac{(-.37)}{3.4} = 1.009$$

The closeness of these values implies that there is a double root near $x = 1$.

Choosing $x_1 = 1.01$ for the next approximation, we get

$$x_1 - 2 \frac{f(x_1)}{f'(x_1)} = 1.01 - 2 \times \frac{0.0002}{0.0403} = 1.0001$$

and

$$x_1 - (2 - 1) \frac{f'(x_1)}{f''(x_1)} = 1.01 - \frac{.0403}{4.06} = 1.0001$$

This shows that there is a double root at $x = 1.0001$ which is quite near the actual root $x = 1$.

Example 2: The equation

$$f(x) = x^3 - 7x^2 + 16x - 12 = 0$$

has a double root at $x = 2$. Starting with the initial approximation $x_0 = 1$, find the root correct to 3 decimal places using the modified Newton-Raphson method with $m = 2$.

Sol. The modified Newton-Raphson method with $m = 2$ becomes,

$$x_{n+1} = x_n - 2 \left[\frac{x_n^3 - 7x_n^2 + 16x_n - 12}{3x_n^2 - 14x_n + 16} \right], n = 0, 1, \dots$$

Starting with $x_0 = 1$, we get

$$x_1 = 1.8$$

$$x_2 = 1.984615385$$

$$x_3 = 1.999884332$$

$$x_4 = 2.000000161$$

$$x_5 = 2.000000161$$

$\therefore$ The root correct to 3 decimal places is 2.000.

Real life/industrial Applications of solution of equations

1. Crime solving Problems:

If somebody is killed, at different time the corpse's temperature is measured and the temperature of room is also recorded. Then we need to determine the time of death from the information we have at hand. Applying the Newton's law of cooling and simplifying, the time of death, t_d is determined by the equation

$$a + t_d - \frac{1}{k} + \left(18 + \frac{1}{k}\right)e^{-kt_d} = b$$

Where a, b are known and k is constant of proportionality in Newton's law of cooling.

2. Volume of a gas:

The van der Waals equation

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT,$$

which relates the pressure (P), volume (V) and temperature (T) of a gas. Here n represents the number of moles of gas present and R is the universal gas constant. The term involving the parameter a corrects the pressure for intermolecular attractive forces, while the term involving the parameter b is a correction for that portion of the volume of the gas that is not compressible due to the intrinsic volume of the gas molecules. What is the volume of gas?

3. Depth of submersion:

Suppose we want to determine how far a spherical object of radius R will sink into a fluid such as water or oil. Given the values for R , ρ_0 and ρ_f the depth to which the object sinks is determined by solving the equation

$$\frac{\rho_f}{3} h^3 - R \rho_f h^2 + \frac{4}{3} R^3 \rho_0 = 0$$

for h where ρ_f is the density of fluid, ρ_0 is mass density of object.

4. Planck's constant:

Planck used his model to explain the spectrum of radiation coming out of a blackbody. There was a well-known UV catastrophe which was that the classical theory predicted that an infinite amount of energy should be emitted, with more and more energy being radiated at shorter wavelengths/higher frequencies. What was observed, was obviously different, in fact, the amount of energy emitted by short distance and long distance wave lengths went to zero and there was a wave length that corresponded to the maximum power radiated. This was observed to obey what was known as Wien's displacement law. By positing that electromagnetic radiation at high frequencies fell off exponentially Planck's law at high energies fundamentally introduced a new constant. Planck could measure Planck's constant immediately by deriving Wien's displacement law which gave

$$h = xkT\lambda_{\max}/c$$

where x is the solution to the transcendental equation

$$e^{-x} = (1 - x/5)$$

which is $x=4.965$. He knew pairs of T and $\lambda_{\max}$ and could immediately measure this constant.

5. Junction Diode Circuits:

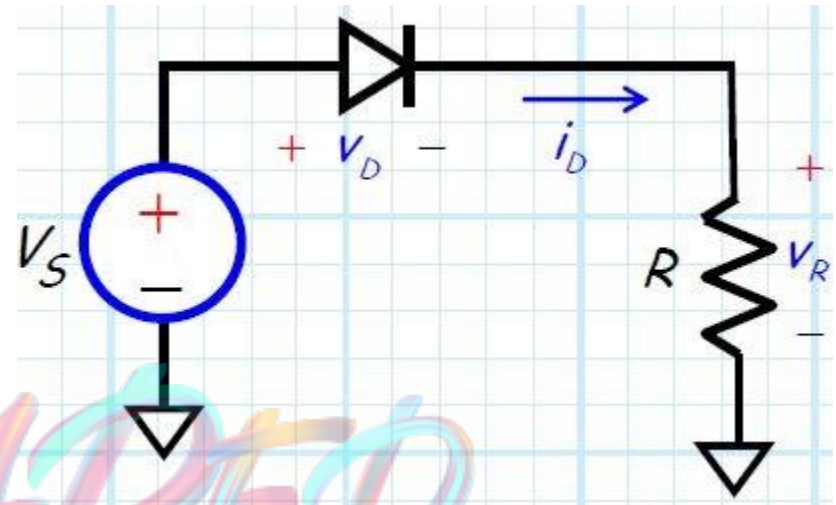
Consider the simple junction diode circuit:

From Kirchhoff's Voltage Law,

$$V_s - v_D - v_R = 0, \text{ Hence,}$$

$$V_s - v_D - Ri_D = 0.$$

$$\text{Hence, } i_D = (V_s - v_D)/R.$$



Likewise from the junction Diode circuit:

$$i_D = I_s \left(e^{v_D/nV_T} - 1 \right)$$

Equating these two equations, we have a single equation with single variable (v_D).

$$\frac{V_s - v_D}{R} = I_s \left(e^{v_D/nV_T} - 1 \right)$$

which is a transcendental equation.

Assignment

1. Use the secant method to determine the root of the equation $\cos x - xe^x = 0$ correct to four decimal places.
2. Show that Newton-Raphson method has quadratic rate of convergence. Using this method find a real root of $2x - \log_{10} x = 7$ correct upto four decimal places.
3. Show that equation $f(x) = 1 - xe^{1-x} = 0$ has a double root at $x = 1$. Find this root using modified Newton-Raphson method taking $x_0 = 0$.

CodeCampus

Unconstrained One Variable Minimization

In golden section search and Fibonacci search method, it is assumed that f is unimodal within the interval i.e. f has only one minimum in the interval.

A f is said to be unimodal in the interval I if $\exists$ an $\alpha^* \in I$ such that α^* minimizes f on I , and for any two points $\alpha_1, \alpha_2 \in I$ such that $\alpha_1 < \alpha_2$, we have

$$\alpha_2 \leq \alpha^* \Rightarrow f(\alpha_1) > f(\alpha_2)$$

$$\alpha_1 \geq \alpha^* \Rightarrow f(\alpha_2) > f(\alpha_1)$$



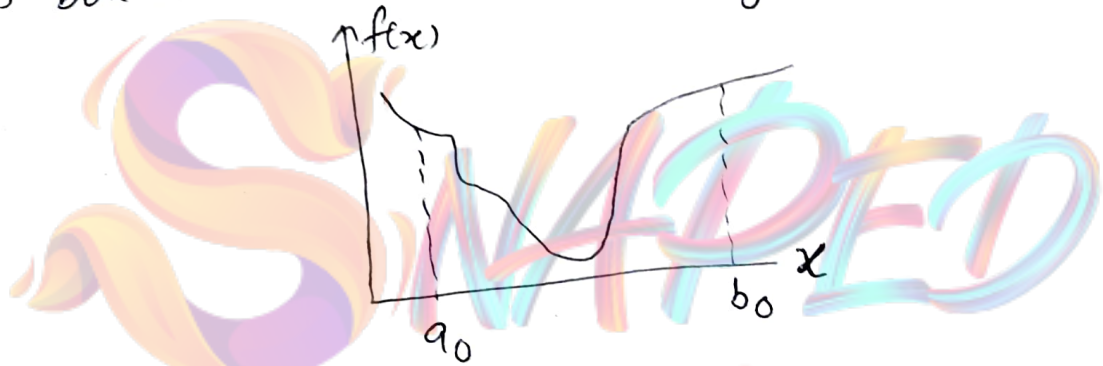
A typical unimodal function

1 Golden^{Section} Search Method:

Determination of the minimizer of a
 $f(x)$ over a closed interval say $[a_0, b_0]$.

The method is based on evaluating the objective function at different points in the interval. We choose these points in such a way that an approximation to the minimizer may be achieved in as few evaluations as possible.

Narrow the range progressively until the minimizer is 'boxed in' within sufficiently accuracy.



Procedure:

Step 1. Choose two points a_1 & b_1 in interval $[a_0, b_0]$

such that

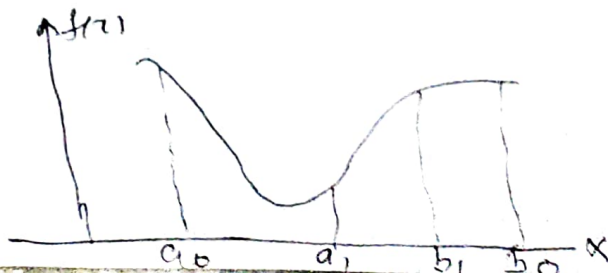
$$a_1 = a_0 + 0.382(b_0 - a_0) \quad \& \quad b_1 = b_0 - 0.382(b_0 - a_0)$$

$$\begin{array}{c} \text{+} \quad \text{---} \quad 0.618 \quad \text{---} \quad \text{---} \quad 0.382 \quad \text{---} \quad \text{+} \\ a_0 \quad \quad a_1 \quad \quad b_1 \quad \quad b_0 \end{array} \quad \text{or} \quad \begin{array}{l} b_1 = a_0 + 0.618(b_0 - a_0) \\ e = 0.382 \\ 1 - e = 0.618 \end{array}$$

Step 2. Check $f(a_1)$ & $f(b_1)$.

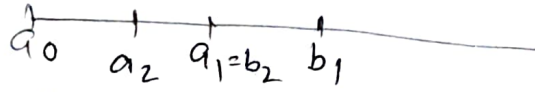
i) If $f(a_1) < f(b_1)$, then minimizer must lie in $[a_0, b_1]$

ii) If $f(a_1) \geq f(b_1)$, then minimizer must lie in $[a_1, b_0]$.



Step 3. Suppose $f(a_1) < f(b_1)$. Then from step 2, $x^* \in [a_0, b_1]$

As $a_1 \in [a_0, b_1]$ and $f(a_1)$ is known, take $b_2 = a_1$.
Thus, only one new evaluation of f at a_2 would be necessary.



Step 4. Repeat the procedure until the interval is reduced to a required accuracy.

Step 5. Choose the mid point of the interval obtained in step 4 and calculate the value of f_n at this point to get the minimum value.

Note: The interval is reduced by the ratio 0.61803 at every stage. Hence, n steps of reduction using the golden section method reduces the range by the factor $(0.61803)^n$.

Ex.1. Use the golden section search to find the value of x that minimizes $f(x) = x^4 - 14x^3 + 60x^2 - 70x$ in the range $[0, 2]$, locate this value of x to within a range of 0.3.

Soln: After N stage, the range is reduced by $(0.61803)^N$. So we choose N so that

$$(0.61803)^N \leq 0.3/2$$

$N = 4$ will do it.

Hence, we need to perform 4 iterations.

Iteration 1. We find a_1 & b_1 by the formulae

$$a_1 = a_0 + 0.382(b_0 - a_0) = 0 + 0.382(2 - 0) = 0.764$$

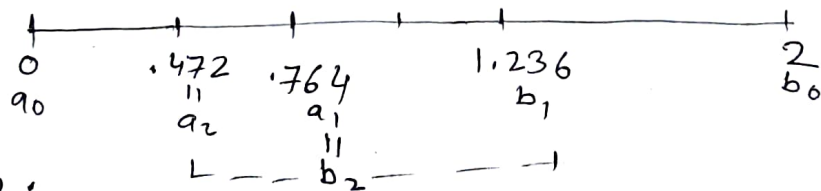
$$\therefore f(a_1) = (0.764)^4 - 14(0.764)^3 + 60(0.764)^2 - 70(0.764)$$

$$f(a_1) = -24.36$$

$$b_1 = b_0 - 0.382(b_0 - a_0) = 1.236$$

$$f(b_1) = -18.96$$

As $f(a_1) < f(b_1)$, interval is reduced to $[a_0, b_1]$
i.e. $[0, 1.236]$



Iteration 2:

$$a_2 = a_0 + 0.382(b_1 - a_0) = 0.472$$

$$f(a_2) = -21.10$$

$$f(b_2) = -24.36$$

$f(b_2) < f(a_2)$, the interval reduces to $[0.472, 1.236]$

Iteration 3:

$$\begin{array}{c}
 \begin{array}{ccc}
 & b_3 & \\
 & 0.944 & \\
 \hline
 0.472 & 0.764 & 1.236 \\
 a_2 & b_2 & b_1 \\
 & a_3 &
 \end{array} \\
 \text{Set } a_3 = b_2 \text{ and compute } b_3 = b_1 - 0.382(b_1 - a_2) \\
 = 1.236 - 0.382 \times 0.764 \\
 \Rightarrow \boxed{b_3 = 0.944} \\
 f(a_3) = -24.36 \\
 f(b_3) = -23.59
 \end{array}$$

$\therefore f(a_3) < f(b_3)$. Hence, interval reduces to $[a_2, b_3]$
 $[0.472, 0.944]$

Iteration 4:

$$\begin{array}{c}
 \begin{array}{cccc}
 & & 0.653 & \\
 & & a_4 & \\
 \hline
 0.472 & & 0.764 & 0.944 \\
 a_2 & & a_3 & b_3 \\
 & & a_4 & \\
 & & b_4 &
 \end{array} \\
 \text{Set } b_4 = a_3 \text{ and calculate } a_4 = a_2 + 0.382(b_3 - a_2) \\
 \therefore a_4 = 0.653 \\
 f(a_4) = -23.84 \\
 f(b_4) = -24.36
 \end{array}$$

$f(b_4) < f(a_4)$. Hence, the interval reduces to
 $[a_4, b_3] = [0.653, 0.944]$.

Thus, the value of x that minimizes f is located in the interval $[0.653, 0.944]$

Note that $b_3 - a_4 = 0.291 < 0.3$.

Ex.2. Use the golden section search to find the value of x that minimizes $f(x) = x(x-3)$ on the interval $0 \leq x \leq 2$ within a range of 0.2.

Ex.3. Minimize the function $f(x) = 4x^3 + x^2 - 7x + 14$ within the interval $[0, 1]$ using golden section method.

2. Fibonacci Search Method:

Fibonacci sequence $\{F_n\}$.

Let $F_0 = 1, F_1 = 1$. Define $F_{k+1} = F_k + F_{k-1}$. Then

$$F_2 = F_1 + F_0 = 1 + 1 = 2, \quad F_3 = F_2 + F_1 = 2 + 1 = 3, \text{ etc.}$$

The sequence is

$$F_1, F_2, F_3, \dots \quad \text{i.e. } 1, 2, 3, 5, 8, 13, \dots$$

which is called as Fibonacci sequence.

Method :

Fibonacci search method is used to locate the minimizer of a function. In golden search method, we have to calculate single value, and ϵ was fixed to 0.382, while in Fibonacci search method, two calculations are required in each stage: one the functional value and other the value of ϵ for each stage.

Calculation of ϵ :

$$\epsilon_1 = 1 - \frac{F_N}{F_{N+1}}, \quad \epsilon_2 = 1 - \frac{F_{N-1}}{F_N}, \quad \epsilon_3 = 1 - \frac{F_{N-2}}{F_{N-1}} \dots$$

$$\dots \quad \epsilon_N = 1 - \frac{F_1}{F_2}$$

Note: 1. The interval (uncertainty range) is reduced by the factor

$$(1 - \epsilon_1)(1 - \epsilon_2) \dots (1 - \epsilon_N) = \frac{1}{F_{N+1}}$$

2. Reduction factor is less than that of golden section method.

3. There is an anomaly in the final iteration because $\epsilon_N = 1 - \frac{F_1}{F_2} = \frac{1}{2}$.

To overcome this situation, a small number ϵ is used and reduction function in the uncertainty range for the Fibonacci method is taken to be as $\frac{1+2\epsilon}{F_{N+1}}$.

Ex! Consider the function $f(x) = x^4 - 14x^3 + 60x^2 - 70x$. Use the Fibonacci search method to find the value of x that minimizes f over the range $[0, 2]$. Locate this value of x to within the range 0.3.

Soln.: In Fibonacci search method, after N steps the range is reduced by $(1+2\epsilon)/F_{N+1}$ in the worst case.

Hence, we need to choose N such that

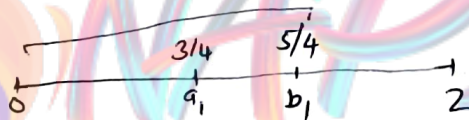
$$\frac{1+2\epsilon}{F_{N+1}} \leq \frac{\text{final range}}{\text{initial range}} = \frac{0.3}{2-0} = 0.15$$

$$\Rightarrow F_{N+1} \geq \frac{1+2\epsilon}{0.15}$$

Choose $\epsilon \leq 0.1$, then we get $N=4$.

Iteration I. Start with

$$1 - \rho_1 = \frac{F_4}{F_5} = \frac{5}{8} \Rightarrow \rho_1 = 1 - \frac{5}{8} = \frac{3}{8}$$



$$\text{Compute } a_1 = a_0 + \rho_1(b_0 - a_0) = 0 + \frac{3}{8} \times (2 - 0) = \frac{3}{4}$$

$$b_1 = b_0 - \rho_1(b_0 - a_0) = 2 - \frac{3}{8} \times 2 = \frac{5}{4}$$

$$f(a_1) = -24.34, \quad f(b_1) = -18.65$$

$$\therefore f(a_1) < f(b_1), \text{ the interval reduces to } [a_0, b_1] = [0, 5/4]$$

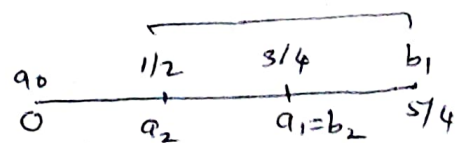
Iteration II

$$1 - \rho_2 = \frac{F_3}{F_4} = \frac{3}{5} \Rightarrow \rho_2 = \frac{2}{5}$$

$$\text{Set } b_2 = a_1 = \frac{3}{4} \text{ and}$$

$$\text{Calculate } a_2 = a_0 + \rho_2(b_1 - a_0)$$

$$= 0 + \frac{2}{5} \times \frac{5}{4} = \frac{1}{2}$$

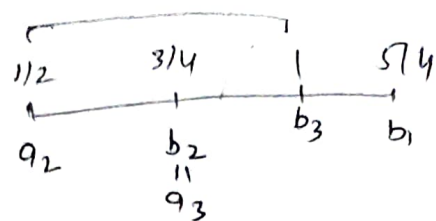


$$\therefore f(a_2) = -21.69. \text{ Also, } f(b_2) = f(a_1) = -24.34$$

$$f(a_2) > f(b_2)$$

$$\text{So, interval is reduced to } [a_2, b_1] = \left[\frac{1}{2}, \frac{5}{4}\right]$$

Iteration 11: $1 - p_3 = \frac{F_2}{F_3} = \frac{2}{3} \Rightarrow p_3 = \frac{1}{3}$



Set $a_3 = b_2 = \frac{3}{4}$

Calculate $b_3 = b_1 - p_3(b_1 - a_2)$

$$= \frac{5}{4} - \frac{1}{3} \times \left(\frac{5}{4} - \frac{1}{2} \right) = \frac{5}{4} - \frac{1}{3} \times \frac{1}{4} = 1.$$

$f(a_3) = f(b_2) = -24.34$

$f(b_3) = -23.$

$f(a_3) < f(b_3)$

$\therefore$ Interval is ~~changed~~ reduced to $[a_2, b_3] = \left[\frac{1}{2}, 1 \right]$

Iteration 12 choose $\epsilon = 0.05$. Then

$1 - p_4 = \frac{F_1}{F_2} = \frac{1}{2}$

Set $b_4 = a_3 = \frac{3}{4}$

Calculate $a_4 = a_2 + (b_1 - a_2)(1 - p_4) = 0.725$

$f(a_4) = -24.27$

$f(b_4) = f(a_3) = -24.34$

$\therefore f(a_4) > f(b_4)$, the interval reduces to $[a_4, b_3] = [0.725, 1]$

(Note that $b_3 - a_4 = 0.275 < 0.3$.)

Hence, the minimizer is located in $[0.725, 1]$.

Ex2. Use Fibonacci search method to find the value of x that minimizes $f(x) = x(x-3)$ on the interval $0 \leq x \leq 2$ within a range of 0.2.

③ Newton's Method:

In the problem of minimizing a function f of a single variable x , assume that at each measurement point $x^{(k)}$, we can calculate $f(x^{(k)})$, $f'(x^{(k)})$ and $f''(x^{(k)})$. We can fit a quadratic function through $x^{(k)}$ that matches its first and second derivative with that of the function f .

$$q(x) = f(x^{(k)}) + f'(x^{(k)})(x - x^{(k)}) + \frac{1}{2} f''(x^{(k)})(x - x^{(k)})^2$$

Note that $q(x^{(k)}) = f(x^{(k)})$, $q'(x^{(k)}) = f'(x^{(k)})$ & $q''(x^{(k)}) = f''(x^{(k)})$

Instead of minimizing f , we minimize its approximation q .

The first order necessary condition for a minimizer of q yields

$$0 = q'(x) = f'(x^{(k)}) + f''(x^{(k)})(x - x^{(k)})$$

Setting $x = x^{(k+1)}$, we get

$$f'(x^{(k)}) + f''(x^{(k)})(x^{(k+1)} - x^{(k)}) = 0$$

$$\Rightarrow x^{(k+1)} - x^{(k)} = - \frac{f'(x^{(k)})}{f''(x^{(k)})}$$

$$\Rightarrow \boxed{x^{(k+1)} = x^{(k)} - \frac{f'(x^{(k)})}{f''(x^{(k)})}, \quad k = 0, 1, 2, 3, \dots}$$

Note: Newton's method works well if $f''(x) > 0$ everywhere. However, if $f''(x) < 0$ for some x , Newton's method may fail to converge to the minimizer.

Ex. Using Newton's method, find the minimizer of $f(x) = \frac{1}{2}x^2 - \sin x$, the initial value is $x^{(0)} = 0.5$. The required accuracy is $\epsilon = 10^{-5}$ in the sense that we stop when $|x^{(k+1)} - x^{(k)}| < \epsilon$.

Soln.:

$$f(x) = \frac{1}{2}x^2 - \sin x$$

$$\therefore f'(x) = x - \cos x \quad \& \quad f''(x) = 1 + \sin x.$$

using Newton's formula

$$x^{(k+1)} = x^{(k)} - \frac{f'(x^{(k)})}{f''(x^{(k)})}, \quad k = 0, 1, 2, \dots$$

$$x^{(1)} = x^{(0)} - \frac{f'(x^{(0)})}{f''(x^{(0)})} = 0.5 - \frac{0.5 - \cos 0.5}{1 + \sin 0.5} = 0.5 - \frac{-0.3775}{1.479} = \underline{\underline{0.7552}}$$

$$x^{(2)} = x^{(1)} - \frac{f'(x^{(1)})}{f''(x^{(1)})} = 0.7552 - \frac{0.02710}{1.685} = \underline{\underline{0.7391}}$$

$$x^{(3)} = 0.7391 - \frac{9.461 \times 10^{-5}}{1.673} = \underline{\underline{0.7390}}$$

$$x^{(4)} = 0.7390 - \frac{1.17 \times 10^{-9}}{1.673} = \underline{\underline{0.7390}} \quad |x^{(4)} - x^{(3)}| < \epsilon = 10^{-5}$$

$$f'(x^{(4)}) = -8.6 \times 10^{-6} \approx 0 \quad \& \quad f''(x^{(4)}) = 1.763 > 0.$$

$\therefore$ We can assume that $x^* \approx x^{(4)}$ is a strict minimizer.

Multivariate Function Minimization

Method of Steepest Descent

Let $f(x) = f(x_1, x_2, \dots, x_n)$ is a fn of variables $x_1, x_2, \dots, x_n$, then $\text{grad } f$ denoted by $\nabla f(x)$ and is obtained by

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \frac{\partial f}{\partial x_3}, \dots, \frac{\partial f}{\partial x_n} \right)$$

In steepest descent method which is an iterative method, iterations are obtained by

$$x^{k+1} = x^k - \alpha^k \nabla f(x^k)$$

where α^k is a parameter called the optimal step size

The parameter α is determined by minimizing

$$f[x^k - \alpha \nabla f(x^k)]$$

which is a function of variable α only.

Procedure:

Step 1. Find $\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)$

Step 2. choose initial approximation x^0 if not already given

Step 3. Calculate $\nabla f(x^0)$.

Step 4. Find $x^1 = x^0 - \alpha^0 \nabla f(x^0)$.

Step 5. Minimize $f(x^1)$ to get the value of α

Step 6. Substitute this value of α in x^1 obtained in step 4.

Step 7. Repeat steps 3 - 6 to obtain next approximation.

Step 8. Repeat step 7 until a required accuracy is obtained.

Ex.1. Find minimum value of $f(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2$ using steepest descent method such that the error does not exceed by 0.05 for function Taking starting point as $x_1 = (1, 1/2)$

Soln. Initial point $x_1 = (1, 1/2)$

$$f(x) = f(x_1, x_2) = x_1^2 + x_1 x_2 + x_2^2 \quad \text{--- ①}$$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2} \right) = (2x_1 + x_2, x_1 + 2x_2)$$

$$\therefore \nabla f(x_1) = (2 \times 1 + 1/2, -1 + 2 \times 1/2) = (3/2, 0)$$

$$\therefore x_2 = x_1 - \gamma_1 \nabla f(x_1) = (\cancel{1}, \cancel{1/2}) - \gamma_1$$

$$= (1, 1/2) - \gamma_1 (3/2, 0)$$

$$= (1 - \frac{3}{2}\gamma_1, \frac{1}{2})$$

$$f(x_2) = (1 - \frac{3}{2}\gamma_1)^2 - (1 - \frac{3}{2}\gamma_1) \left(\frac{1}{2} \right) + \left(\frac{1}{2} \right)^2$$

$$= 1 - 3\gamma_1 + \frac{9}{4}\gamma_1^2 - \frac{1}{2} + \frac{3}{4}\gamma_1 + \frac{1}{4}$$

$$= \frac{9}{4}\gamma_1^2 - \frac{5}{4}\gamma_1 + \frac{3}{4}$$

γ_1 is obtained by minimize $f(x_2)$

$$\frac{df}{d\gamma_1} = \frac{18}{4}\gamma_1 - \frac{5}{4} = 0 \Rightarrow \boxed{\gamma_1 = \frac{1}{2}}$$

$$\therefore x_2 = \left(1 - \frac{3}{2} \cdot \frac{1}{2}, \frac{1}{2} \right) = (1/4, 1/2)$$

$$\therefore f(x_2) = f(x_1, x_2) = \left(\frac{1}{4} \right)^2 - \frac{1}{4} \times \frac{1}{2} + \frac{1}{4} = 3/16 \quad [\text{From ①}]$$

$$f(x_1) = f(1, 1/2) = 1^2 - 1 \times \frac{1}{2} + \frac{1}{4} = 3/4$$

$$\text{Now, } |f(x_2) - f(x_1)| = \left| \frac{3}{16} - \frac{3}{4} \right| = \left| 1 - \frac{9}{16} \right| = \frac{9}{16} > 0.05$$

$\therefore$ We need to proceed to find x_3

$$x_3 = x_2 - \gamma_2 \nabla f(x_2)$$

$$x_3 = \left(\frac{1}{4}, \frac{1}{2}\right) - r_2 \left(2 \times \frac{1}{4} - \frac{1}{2}, -\frac{1}{4} + 2 \times \frac{1}{2}\right) \\ = \left(\frac{1}{4}, \frac{1}{2}\right) - r_2 \left(0, \frac{3}{4}\right)$$

$$\Rightarrow x_3 = \left(\frac{1}{4}, \frac{1}{2} - \frac{3}{4}r_2\right)$$

$$f(x_3) = \left(\frac{1}{4}\right)^2 - \frac{1}{4} \left(\frac{1}{2} - \frac{3}{4}r_2\right) + \left(\frac{1}{2} - \frac{3}{4}r_2\right)^2$$

$$= \frac{1}{16} - \frac{1}{8} + \frac{3}{16}r_2 + \frac{1}{4} - \frac{3}{4}r_2 + \frac{9}{16}r_2^2 = \frac{9}{16}r_2^2 - \frac{9}{16}r_2 + \frac{3}{16}$$

$$= \frac{9}{16}r_2^2 - \frac{9}{16}r_2 + \frac{3}{16}$$

r_2 is obtained by minimizing $f(x_3)$

$$\frac{df}{dr_2} = \frac{18}{16}r_2 - \frac{9}{16} = 0 \Rightarrow r_2 = \frac{9}{18} = \frac{1}{2}$$

$$\Rightarrow r_2 = \frac{1}{2}$$

$$\therefore x_3 = \left(\frac{1}{4}, \frac{1}{2} - \frac{3}{4} \times \frac{1}{2}\right) = \left(\frac{1}{4}, \frac{1}{8}\right)$$

$$\therefore f(x_3) = \frac{9}{16} \times \frac{1}{4} - \frac{9}{16} \times \frac{1}{2} + \frac{3}{16} = \frac{9 - 18 + 12}{64} = \frac{3}{64}$$

$$\text{Now, } |f(x_3) - f(x_2)| = \left|\frac{3}{64} - \frac{3}{16}\right| = \left|-\frac{9}{64}\right| = \frac{9}{64} \neq 0.05$$

$\therefore$ Calculate x_4 .

$$x_4 = x_3 - r_3 \nabla f(x_3) = \left(\frac{1}{4}, \frac{1}{8}\right) - r_3 \left(2 \times \frac{1}{4} - \frac{1}{8}, -\frac{1}{4} + 2 \times \frac{1}{8}\right) \\ = \left(\frac{1}{4}, \frac{1}{8}\right) - r_3 \left(\frac{3}{8}, 0\right) = \left(\frac{1}{4} - \frac{3}{8}r_3, \frac{1}{8}\right)$$

$$f(x_4) = \left(\frac{1}{4} - \frac{3}{8}r_3\right)^2 - \left(\frac{1}{4} - \frac{3}{8}r_3\right) \times \frac{1}{8} + \left(\frac{1}{8}\right)^2$$

$$= \frac{1}{16} - \frac{3}{16}r_3 + \frac{9}{64}r_3^2 - \frac{1}{32} + \frac{3}{64}r_3 + \frac{1}{64} = \frac{9}{64}r_3^2 - \frac{9}{64}r_3 + \frac{3}{64}$$

r_3 is obtained by minimizing $f(x_4)$

$$\frac{df}{dr_3} = \frac{18}{64}r_3 - \frac{9}{64} = 0 \Rightarrow r_3 = \frac{1}{2}$$

$$\therefore x_4 = \left(\frac{1}{4} - \frac{3}{8} \times \frac{1}{2}, \frac{1}{8}\right) = \left(\frac{1}{16}, \frac{1}{8}\right)$$

$$f(x_4) = \frac{9}{64} \times \frac{1}{4} - \frac{9}{64} \times \frac{1}{2} + \frac{3}{64} = \frac{3}{256}$$

$$\text{Now, } |f(x_4) - f(x_3)| = \left|\frac{3}{256} - \frac{3}{64}\right| = \left|-\frac{9}{256}\right| = \frac{9}{256} = 0.035 < 0.05$$

$\therefore$ Min. value of f_n is $\boxed{3/256}$ and point is $\boxed{(1/16, 1/8)}$

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Ex.2 Using steepest descent method, minimize

$$f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$$

starting from the point $x_1 = (0, 0)$.

(Hint: In such case, perform the iterations until $\nabla f(x_i) \cong 0$.)

$$x_2 = (-1, 1)$$

$$x_3 = (-0.8, 1.2)$$

$$x_4 = (-1.0, 1.4)$$

$$x_5 = (-0.96, 1.44)$$

$$\therefore \nabla f(x_5) = (0.04, -0.04) \cong (0, 0)$$

$\therefore x_5$ is optimum

Ex.3 Perform three iterations of the steepest descent method to minimize

$$f(x_1, x_2) = 4x_1^2 + 6x_2^2 - 8x_1x_2 \text{ with}$$

initial point $x_1 = (2, 1)$.